# Lesson 1: Mixed and Entire Radicals

*Evaluating roots exactly, rewriting radicals in simplest mixed form and as entire radicals, and using entire radicals to compare sizes.*

## 1. Roots and Radicals

A **radical** is an expression of the form $\sqrt[n]{x}$, where $n$ is a natural number greater than 1. The number $n$ is the **index** and $x$ is the **radicand**. When no index is written, the index is 2 (a square root). For example, in $\sqrt[3]{40}$ the index is 3 and the radicand is 40.

- Every positive number has two square roots: 36 has square roots 6 and $-6$. The symbol $\sqrt{\ }$ means only the **principal** (non-negative) root: $\sqrt{36} = 6$, and $-\sqrt{36} = -6$.
- A negative number has **no real** square root (or fourth root, or any even root): (any real number)$^2 \geq 0$.
- Every real number has exactly one cube root (or fifth root, or any odd root): $\sqrt[3]{-64} = -4$ because $(-4)^3 = -64$.

**Even index** ($\sqrt{x}$, $\sqrt[4]{x}$, ...): the radicand must be $\geq 0$, and the root is $\geq 0$. **Odd index** ($\sqrt[3]{x}$, $\sqrt[5]{x}$, ...): any radicand; $\sqrt[3]{-x} = -\sqrt[3]{x}$.

A **rational number** can be written as a fraction $\frac{a}{b}$ of integers ($b \neq 0$); its decimal ends or repeats. An **irrational number** cannot; its decimal goes on forever without repeating. The root of a perfect square, cube, etc. is rational ($\sqrt{0.25} = 0.5$, $\sqrt[3]{\frac{8}{27}} = \frac{2}{3}$). Any other root is irrational ($\sqrt{2} = 1.414\,213\ldots$). Together, the rational and irrational numbers make up the **real numbers**.

### Example 1

Without a calculator, evaluate where possible.

(a) $\sqrt{121}$ (b) $-\sqrt{49}$ (c) $\sqrt{-49}$

(d) $\sqrt[3]{-125}$ (e) $\sqrt[4]{81}$ (f) $\sqrt[4]{-81}$

(g) $\sqrt[5]{-1}$ (h) $\sqrt[3]{\frac{8}{27}}$ (i) $\sqrt{0.09}$

### Example 2

(a) Classify each number as rational or irrational.

(i) $\sqrt{50}$ (ii) $\sqrt[3]{-216}$ (iii) $\sqrt{\frac{49}{64}}$

(iv) $\sqrt[4]{40}$ (v) $\sqrt{0.4}$ (vi) $\sqrt{0.49}$

(b) Between which two consecutive whole numbers does each number lie? (i) $\sqrt{50}$ (ii) $\sqrt[3]{50}$

### 2. Multiplying and Dividing Under One Root

Roots with the **same index** can be multiplied or divided under one radical sign. Check: $\sqrt{4} \times \sqrt{9} = 2 \times 3 = 6$ and $\sqrt{4 \times 9} = \sqrt{36} = 6$. This does **not** work for addition or subtraction: $\sqrt{9} + \sqrt{16} = 3 + 4 = 7$, but $\sqrt{9+16} = \sqrt{25} = 5$.

$\sqrt[n]{a} \times \sqrt[n]{b} = \sqrt[n]{ab}$ and $\dfrac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\dfrac{a}{b}}$ ($b \neq 0$; when $n$ is even, $a \geq 0$ and $b \geq 0$)

In general, $\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$ and $\sqrt{a-b} \neq \sqrt{a} - \sqrt{b}$.

**Example 3**

State whether each statement is true or false. Show a check.

(a) $\sqrt{2} \times \sqrt{18} = 6$

(b) $\sqrt{36+64} = \sqrt{36} + \sqrt{64}$

(c) $\dfrac{\sqrt{72}}{\sqrt{2}} = 6$

(d) $\sqrt{169} - \sqrt{25} = \sqrt{144}$

(e) $\sqrt[3]{4} \times \sqrt[3]{16} = 4$

(f) $\sqrt[3]{2} + \sqrt[3]{6} = \sqrt[3]{8}$

### 3. Entire Radicals to Mixed Radicals

An **entire radical** has its whole value under the radical sign: $\sqrt{48}$, $\sqrt[3]{54}$. A **mixed radical** has a number in front: $4\sqrt{3}$, $3\sqrt[3]{2}$ (the number multiplies the radical). A radical is in **simplest form** when the radicand has no perfect-square factor (index 2), no perfect-cube factor (index 3), and so on.

To simplify, write the radicand as a product using the **largest** perfect power that divides it, split the root, and take that root out:

$$\sqrt{48} = \sqrt{16 \times 3} = \sqrt{16} \times \sqrt{3} = 4\sqrt{3} \qquad \sqrt[3]{54} = \sqrt[3]{27 \times 2} = \sqrt[3]{27} \times \sqrt[3]{2} = 3\sqrt[3]{2}$$

**Perfect squares:** 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225, ...
**Perfect cubes:** 8, 27, 64, 125, 216, 343, 512, 729, 1000, ... **Perfect fourth powers:** 16, 81, 256, 625, ...

**Example 4**

Write each radical as a mixed radical in simplest form.

(a) $\sqrt{75}$

(b) $\sqrt{120}$

(c) $\sqrt{252}$

(d) $3\sqrt{98}$

(e) $\sqrt[3]{250}$

(f) $\sqrt[3]{-432}$

(g) $\sqrt[4]{162}$

(h) $-2\sqrt[3]{375}$

(i) $\sqrt[4]{1280}$

For a large radicand, use its **prime factorization**. For a square root, each *pair* of equal factors comes out as one factor; for a cube root, each *triple*.

**Example 5**

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(a) Simplify $\sqrt{2700}$ using prime factors.

(b) Simplify $\sqrt[3]{3024}$ using prime factors.

(c) Kai wrote $\sqrt{72} = 2\sqrt{18}$. Is this in simplest form? Explain and finish the simplification.

Simplifying also lets you estimate a radical from one known value. Since 100 and $\frac{1}{100}$ are perfect squares, $\sqrt{300} = 10\sqrt{3}$ and $\sqrt{0.03} = \sqrt{\frac{3}{100}} = \frac{\sqrt{3}}{10}$.

**Example 6**

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Given $\sqrt{3} \approx 1.732$, determine an approximate value of each without a calculator.

(a) $\sqrt{300}$ (b) $\sqrt{12}$ (c) $\sqrt{0.03}$ (d) $\sqrt{48}$

### 4. Mixed Radicals to Entire Radicals

Every mixed radical can be written as an entire radical. Write the number in front as a radical with the same index (raise it to the power of the index), then multiply the radicands:

$$5\sqrt{2} = \sqrt{5^2} \times \sqrt{2} = \sqrt{25 \times 2} = \sqrt{50} \qquad 2\sqrt[3]{7} = \sqrt[3]{2^3} \times \sqrt[3]{7} = \sqrt[3]{56}$$

$a\sqrt[n]{b} = \sqrt[n]{a^n \times b}$
**Negative number in front:** for an *odd* index the sign can go inside ($-2\sqrt[3]{7} = \sqrt[3]{(-2)^3 \times 7} = \sqrt[3]{-56}$); for an *even* index it stays outside ($-5\sqrt{2} = -\sqrt{50}$, because $\sqrt{50}$ is never negative).

**Example 7**

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Write each mixed radical as an entire radical.

(a) $4\sqrt{3}$ (b) $7\sqrt{2}$ (c) $-2\sqrt{11}$ (d) $3\sqrt[3]{4}$

(e) $-5\sqrt[3]{2}$ (f) $2\sqrt[4]{5}$ (g) $\frac{1}{2}\sqrt{20}$ (h) $\frac{2}{3}\sqrt[3]{54}$

### 5. Comparing and Ordering Radicals

To compare radicals without a calculator, write each one as an **entire radical with the same index**. Then the larger radicand gives the larger value. This works for negative cube roots too: $-80 > -81$, so $\sqrt[3]{-80} > \sqrt[3]{-81}$.

**Example 8**

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Without a calculator, arrange the radicals in the order asked.

(a) $3\sqrt{6},\ 2\sqrt{13},\ 5\sqrt{2},\ \sqrt{55},\ 4\sqrt{3}$ from least to greatest.

(b) $2\sqrt[3]{9},\ 3\sqrt[3]{3},\ \sqrt[3]{70},\ -2\sqrt[3]{10},\ -3\sqrt[3]{3}$ from greatest to least.

### 6. Radicals in Context

Lengths found from areas, volumes and the Pythagorean theorem are often irrational. Give them **exactly**, as a radical in simplest form.

| |
|---|
| Square of area $A$: side $= \sqrt{A}$. Cube of volume $V$: edge $= \sqrt[3]{V}$. Right triangle: $c = \sqrt{a^2 + b^2}$. Distance between two points: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ (the Pythagorean theorem with the horizontal and vertical changes as legs). |

**Example 9**

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(a) A square floor has an area of 288 $m^2$. Determine the exact side length and perimeter.

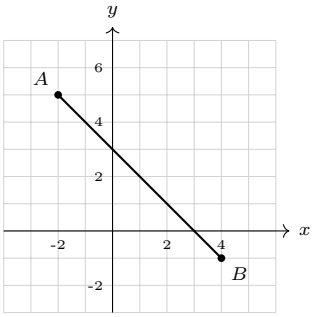
(b) A cube-shaped planter holds 1080 $cm^3$ of soil when full. Determine the exact edge length.

**Example 10**

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(a) A right triangle has legs 4 cm and 8 cm. Determine the exact length of the hypotenuse.

(b) Determine the exact length of segment $AB$, with $A(-2, 5)$ and $B(4, -1)$.



## Lesson 2: Radicals with Variables and Their Restrictions

*Finding the values of the variable for which a radical is defined, then simplifying and converting radicals with variable radicands.*

### 1. Restrictions on the Variable

A radical with a variable radicand, such as $\sqrt{x-6}$, is a real number only for some values of $x$. The values that are allowed are written as a **restriction**.

- **Even index:** the radicand must be $\geq 0$. Set the radicand $\geq 0$ and solve. (Remember: dividing or multiplying both sides of an inequality by a negative number reverses the inequality sign.)
- **Odd index:** there is no restriction; the variable can be any real number.
- If the radical is in a **denominator**, it also cannot equal 0, so the radicand must be $> 0$.

| $\sqrt{\text{radicand}}$, $\sqrt[4]{\text{radicand}}$, …: solve radicand $\geq 0$. | $\sqrt[3]{\text{radicand}}$, $\sqrt[5]{\text{radicand}}$, …: $x \in \mathbb{R}$ (all real numbers). |
|---|---|

**Example 1**

State the restriction on the variable.

(a) $\sqrt{x-6}$

(b) $\sqrt{2x+5}$

(c) $\sqrt{9-3x}$

(d) $\sqrt[3]{x-4}$

(e) $\sqrt[4]{-x}$

(f) $\sqrt{12x}$

**Example 2**

For what values of $x$ is each expression defined? Explain.

(a) $\sqrt{x^2+4}$

(b) $\sqrt{x^2}$

(c) $\sqrt{x^3}$

(d) $\dfrac{8}{\sqrt{x+2}}$

### 2. Simplifying Powers Under a Radical

A power comes out of a root by **dividing its exponent by the index**: $\sqrt{x^8} = x^4$ because $(x^4)^2 = x^8$, and $\sqrt[3]{x^{12}} = x^4$ because $(x^4)^3 = x^{12}$. When the exponent does not divide evenly, the remainder stays inside:

$$\sqrt{x^7} = \sqrt{x^6 \cdot x} = x^3\sqrt{x} \qquad \sqrt[3]{a^{11}} = \sqrt[3]{a^9 \cdot a^2} = a^3\sqrt[3]{a^2}$$

| $\sqrt[n]{x^m}$: divide $m$ by $n$. The **quotient** is the exponent outside; the **remainder** is the exponent left inside. |
|---|

**A caution about even roots.** Try $x = -5$: $\sqrt{x^2} = \sqrt{25} = 5$, which is *not* $x$. In general $\sqrt{x^2} = |x|$ (the absolute value of $x$, studied in Unit 7). To avoid this, for an **even index** we state that the variables are **non-negative** ($x \geq 0$); then

$\sqrt{x^2} = x$. For an odd index no restriction is needed: $\sqrt[3]{x^3} = x$ for every $x$.

**Example 3**

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Simplify. State the restrictions where needed.

(a) $\sqrt{x^8}$ (b) $\sqrt{a^7}$ (c) $\sqrt[3]{y^9}$

(d) $\sqrt[3]{m^{11}}$ (e) $\sqrt[4]{x^9}$ (f) $\sqrt{x^4y^5}$

### 3. Simplifying Radicals with Variable Radicands

Simplify the number and each variable separately, then multiply everything that came out and everything that stayed in:

$$\sqrt{18x^3} = \sqrt{9 \cdot 2 \cdot x^2 \cdot x} = 3x\sqrt{2x}, \quad x \geq 0$$

Unless told otherwise, for an even index state that each variable is $\geq 0$. A negative number under an odd root comes out as negative: $\sqrt[3]{-8x^4} = -2x\sqrt[3]{x}$.

**Example 4**

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Write each radical in simplest form. State the restrictions where needed.

(a) $\sqrt{45a^5}$ (b) $\sqrt{72x^4y^7}$ (c) $\sqrt[3]{-54m^7}$

(d) $\sqrt{98x^2y^3}$ (e) $\sqrt[3]{40a^4b^9}$ (f) $\sqrt[4]{48x^9}$

### 4. Mixed to Entire with Variables

Raise everything in front of the radical to the power of the index, then multiply it into the radicand. As with numbers, a negative sign in front goes inside only for an odd index.

$$3x\sqrt{2x} = \sqrt{(3x)^2 \cdot 2x} = \sqrt{9x^2 \cdot 2x} = \sqrt{18x^3}, \quad x \geq 0$$

**Example 5**

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Write each mixed radical as an entire radical. Assume variables under an even index are $\geq 0$.

(a) $4a\sqrt{a}$ (b) $2x^2\sqrt{5y}$ (c) $-3m\sqrt[3]{2m^2}$

(d) $ab^2\sqrt[3]{4a}$ (e) $-2y\sqrt{y}$ (f) $5x\sqrt{3}$

### 5. Putting It Together

**Example 6**

Give each answer in simplest form, with its restriction.

(a) A square has an area of $48x^6$ cm$^2$. Determine its side length.

(b) A cube has a volume of $54a^4$ m$^3$. Determine its edge length.

(c) A right triangle has legs $3x$ and $6x$. Determine the length of the hypotenuse.

**Example 7**

Consider the expression $\sqrt{20-4x}$.

(a) State the restriction on $x$.

(b) Evaluate the expression exactly, in simplest form, for $x=-5$ and for $x=2$.

(c) Can the expression be evaluated for $x=6$? Explain.

## Lesson 3: Adding and Subtracting Radicals

*Combining like radicals, simplifying each radical first so that like radicals appear, and finding exact perimeters and lengths.*

### 1. Like Radicals

**Like radicals** have the **same index** and the **same radicand**. For example, $4\sqrt{7}$ and $-\sqrt{7}$ are like radicals; $2\sqrt{3}$ and $2\sqrt{5}$ are not (different radicands), and $\sqrt{5}$ and $\sqrt[3]{5}$ are not (different indices).

Add or subtract like radicals the same way you collect like terms in algebra ($4x+3x=7x$): add or subtract the **coefficients** and keep the radical part.

| | |
|---|---|
| $m\sqrt[n]{a}+k\sqrt[n]{a}=(m+k)\sqrt[n]{a}$ | Only like radicals can be combined. Never add or subtract the radicands. |

For example, $\sqrt{9}+\sqrt{16}=3+4=7$, but $\sqrt{9+16}=\sqrt{25}=5$. So $\sqrt{a}+\sqrt{b}\neq\sqrt{a+b}$. Unlike radicals stay as separate terms, just as $2x+5y$ cannot be combined. Rational terms (numbers with no radical) combine only with other rational terms.

**Example 1**

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Simplify by combining like radicals.

(a) $6\sqrt{5}+3\sqrt{5}$

(b) $2\sqrt{13}-8\sqrt{13}+\sqrt{13}$

(c) $7\sqrt[3]{6}-10\sqrt[3]{6}$

(d) $5\sqrt{3}+2\sqrt{7}-\sqrt{3}+4\sqrt{7}$

(e) $9-4\sqrt{2}+3\sqrt{2}-5$

(f) $3\sqrt{2}+2\sqrt[3]{2}$

### 2. Simplify First, Then Combine

Radicals that look unlike may turn into like radicals once each one is in simplest form.

**To add or subtract radicals:** (1) write every radical as a mixed radical in simplest form (Lesson 1); (2) combine the like radicals; (3) leave unlike radicals as separate terms.

A fraction in front of a radical is a coefficient: simplify the radical, then multiply the fraction by the number that came out.

**Example 2**

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Simplify.

(a) $\sqrt{48}+\sqrt{75}$

(b) $3\sqrt{8}-\sqrt{50}+\sqrt{18}$

(c) $\sqrt{20}+\sqrt{27}+\sqrt{45}-\sqrt{12}$

(d) $2\sqrt{72}-3\sqrt{32}+\sqrt{98}$

**Example 3**

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Simplify. Look for perfect cubes $(8, 27, 64, 125, \dots)$ or perfect fourth powers $(16, 81, 256, \dots)$.

(a) $\sqrt[3]{54} + \sqrt[3]{16}$ (b) $2\sqrt[3]{81} - \sqrt[3]{24} + \sqrt[3]{375}$ (c) $\sqrt[3]{-40} + 3\sqrt[3]{135}$

(d) $\sqrt[4]{48} + \sqrt[4]{243}$ (e) $\sqrt{12} + \sqrt[3]{24}$ (f) $\sqrt{50} + \sqrt[3]{16} - \sqrt{8}$

**Example 4**

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Simplify.

(a) $\frac{1}{2}\sqrt{48} + \frac{2}{3}\sqrt{27}$ (b) $\frac{3}{4}\sqrt{32} - \frac{1}{5}\sqrt{50}$ (c) $\frac{2}{3}\sqrt{18} + \frac{1}{4}\sqrt{8}$

### 3. Radicals with Variables

Simplify each radical as in Lesson 2, then combine like radicals. Coefficients that contain variables combine only when they are like terms: $3x\sqrt{2x} + 5x\sqrt{2x} = 8x\sqrt{2x}$. State the restrictions: for an even index, each radicand must be $\geq 0$. A cube root needs no restriction.

**Example 5**

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Simplify. State the restrictions on the variable.

(a) $\sqrt{9a} + \sqrt{25a} - \sqrt{4a}$ (b) $\sqrt{12x^3} + x\sqrt{27x}$

(c) $\sqrt[3]{16y^4} - y\sqrt[3]{54y}$ (d) $\sqrt{4x - 8} + \sqrt{x - 2}$

### 4. Perimeter and Missing Lengths

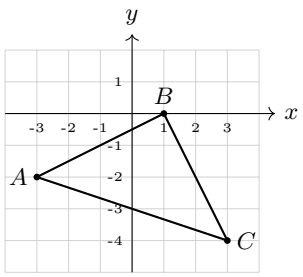
Exact answers are left in simplest radical form. The length of a slanted segment on a grid comes from the Pythagorean theorem: draw the right triangle under it, then

$$\text{length} = \sqrt{(\text{run})^2 + (\text{rise})^2}$$

To find a missing side from a perimeter, subtract the known sides from the perimeter. Put brackets around a side that has two terms before subtracting it.

**Example 6**

Triangle $ABC$ has vertices $A(-3,-2)$, $B(1,0)$ and $C(3,-4)$.

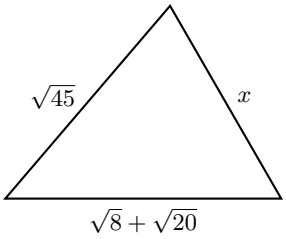



(a) Determine the length of each side in simplest form.

(b) Determine the perimeter of $\triangle ABC$ in simplest form.

**Example 7**

The perimeter of the triangle is $7\sqrt{5}+3\sqrt{2}$ cm. Determine the exact length of side $x$ in simplest form.




**5. Unknown Radical Expressions and Sequences**

An equation such as $y+\sqrt{27}=2\sqrt{75}$ is solved like a linear equation: isolate the unknown, then simplify and combine. Terms with radicals can also form an arithmetic sequence (Unit 1): find $d$ by subtracting consecutive terms, collecting the rational parts and the radical parts separately.

**Example 8**

(a) If $y+\sqrt{27}=2\sqrt{75}$, determine $y$ in simplest form, then simplify $y-\sqrt{12}$.

(b) The sequence $3+\sqrt{2},\ 5+4\sqrt{2},\ 7+7\sqrt{2},\ldots$ is arithmetic. Determine $d$, the next two terms, and $t_{10}$.

## Lesson 4: Multiplying Radicals

*Multiplying coefficients and radicands, expanding products that contain radicals, and using conjugates to get a rational product.*

### 1. Multiplying Radicals with the Same Index

Radicals with the **same index** are multiplied by multiplying the coefficients together and the radicands together. Then write the result in simplest form.

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|---|
| $m\sqrt[n]{a} \cdot k\sqrt[n]{b} = mk\sqrt[n]{ab}$ (same index $n$; $a \geq 0$ and $b \geq 0$ when $n$ is even) |
| $(\sqrt{a})^2 = a$ for $a \geq 0$ $\qquad$ $(\sqrt[3]{a})^3 = a$ |

Two useful habits:

- Simplifying each radical *before* multiplying keeps the numbers small: $\sqrt{12} \cdot \sqrt{27} = 2\sqrt{3} \cdot 3\sqrt{3} = 6 \cdot 3 = 18$.
- The rule needs the same index: $\sqrt{2} \cdot \sqrt[3]{2}$ is *not* $\sqrt{4}$ or $\sqrt[3]{4}$, and it is left as it is in this course.

**Example 1**

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Multiply and simplify.

(a) $\sqrt{6} \cdot \sqrt{10}$

(b) $(3\sqrt{2})(5\sqrt{7})$

(c) $(-2\sqrt{6})(4\sqrt{15})$

(d) $\sqrt{12} \cdot \sqrt{27}$ (two ways)

(e) $(2\sqrt[3]{4})(3\sqrt[3]{6})$

(f) $(\sqrt{5})(-2\sqrt{3})(\sqrt{15})$

**Example 2**

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Simplify.

(a) $\left(\sqrt{7}\right)^2$

(b) $\left(5\sqrt{3}\right)^2$

(c) $-\left(2\sqrt{6}\right)^2$

(d) $\left(2\sqrt[3]{3}\right)^3$

(e) Write $6\sqrt{15}$ as a product of two mixed radicals in two different ways.

### 2. Multiplying Radicals with Variables

The same rule works with variable radicands. Multiply, then remove perfect-square (or perfect-cube) factors, including even powers of the variable. For an even index, every radicand in the question must be $\geq 0$, so state that restriction.

**Example 3**

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Multiply and simplify. State the restrictions on the variables.

(a) $(3\sqrt{x})(5\sqrt{y})$

(b) $(2\sqrt{6a})(\sqrt{10a})$

(c) $\left(\sqrt[3]{4m^2}\right)\left(\sqrt[3]{6m^2}\right)$

(d) $\left(\sqrt{2x^3}\right)\left(3\sqrt{8x}\right)$

### 3. Distributing a Radical

To multiply a radical by an expression in brackets, multiply it by **each term** (the distributive property), then simplify each product and combine like radicals.

**Example 4**

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Expand and simplify. State the restrictions in (c).

(a) $\sqrt{3}\left(\sqrt{6}+4\right)$

(b) $2\sqrt{5}\left(3\sqrt{10}-\sqrt{5}+\sqrt{15}\right)$

(c) $\sqrt{x}\left(\sqrt{x}+3\sqrt{xy}\right)$

### 4. Multiplying Binomials

Multiply each term of the first binomial by each term of the second ("FOIL"), simplify each product, then combine like terms. To square a binomial, write it as a product of two equal binomials first.

| $(a+b)^2 = a^2+2ab+b^2$ $\quad$ $(a-b)^2 = a^2-2ab+b^2$ $\quad$ Never $(a+b)^2 = a^2+b^2$. |
|---|

**Example 5**

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Expand and simplify.

(a) $\left(3+\sqrt{2}\right)\left(5-2\sqrt{2}\right)$

(b) $\left(2\sqrt{3}+\sqrt{5}\right)\left(\sqrt{3}-4\sqrt{5}\right)$

(c) $\left(\sqrt{6}-\sqrt{2}\right)\left(\sqrt{3}+2\sqrt{2}\right)$

**Example 6**

Expand and simplify. State the restrictions in (c).

(a) $\left(\sqrt{5}+3\right)^2$ (b) $\left(2\sqrt{7}-\sqrt{3}\right)^2$ (c) $\left(\sqrt{x}-2\sqrt{y}\right)^2$

### 5. Conjugates

The **conjugate** of $a\sqrt{b}+c\sqrt{d}$ is $a\sqrt{b}-c\sqrt{d}$: same two terms, opposite sign between them. When conjugates are multiplied, the two middle terms are opposites and cancel, and squaring each term removes the radicals:

$$\left(a\sqrt{b}+c\sqrt{d}\right)\left(a\sqrt{b}-c\sqrt{d}\right)=a^2b-c^2d$$ The product of conjugates is **rational**.

A number like $\sqrt{7}$ (the square root of a whole number that is not a perfect square) is **irrational**: its decimal never ends or repeats. This shows that the product of two irrational numbers can be rational. Conjugates are used in Lesson 6 to remove radicals from a denominator.

**Example 7**

Write the conjugate of each expression, then multiply the expression by its conjugate. State the restrictions in (d).

(a) $\sqrt{7}+2$ (b) $3\sqrt{2}-\sqrt{5}$

(c) $4+2\sqrt{6}$ (d) $\sqrt{a}+3\sqrt{b}$

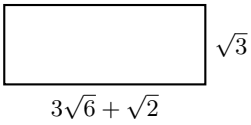
### 6. Area Problems

Area formulas work the same way with radical side lengths: rectangle $A=lw$, square $A=s^2$, triangle $A=\frac{1}{2}bh$. Give the answer exactly, in simplest form.

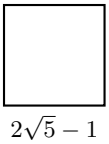
**Example 8**

Determine the area of each figure in simplest form. Dimensions are in centimetres.

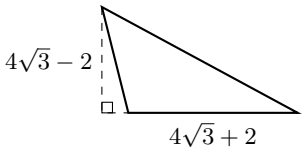
(a) Rectangle




(b) Square




(c) Triangle




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## Lesson 5: Dividing Radicals and Rationalizing Monomial Denominators

*Dividing radicals with the same index, dividing every term of a numerator, and clearing a one-term radical from a denominator.*

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### 1. Dividing Radicals

Radicals with the **same index** can be divided: divide the coefficients, divide the radicands, then write the result in simplest form. This is the multiplication rule $\sqrt{a}\cdot\sqrt{b}=\sqrt{ab}$ from Lesson 4 run backward.

$$\frac{m\sqrt[n]{a}}{k\sqrt[n]{b}}=\frac{m}{k}\sqrt[n]{\frac{a}{b}} \qquad (k\neq 0,\ b\neq 0;\ \text{when } n \text{ is even, } a\geq 0 \text{ and } b>0)$$

**Example 1**

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Divide and write in simplest form.

(a) $\dfrac{\sqrt{48}}{\sqrt{3}}$

(b) $\dfrac{12\sqrt{42}}{4\sqrt{6}}$

(c) $\dfrac{10\sqrt{54}}{5\sqrt{3}}$

(d) $\dfrac{\sqrt[3]{250}}{\sqrt[3]{2}}$

(e) $\dfrac{-18\sqrt[3]{80}}{6\sqrt[3]{5}}$

(f) $\dfrac{6\sqrt{35}}{9\sqrt{5}}$

**Example 2**

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When the radicands do not divide evenly, either reduce the fraction under the radical or write each radical in simplest form first. Simplify.

(a) $\dfrac{\sqrt{98}}{\sqrt{18}}$

(b) $\dfrac{4\sqrt{90}}{3\sqrt{40}}$

(c) $\dfrac{15\sqrt{72}}{10\sqrt{50}}$

(d) $\dfrac{6\sqrt[3]{54}}{\sqrt[3]{16}}$

### 2. Dividing Each Term of a Numerator

A fraction bar groups the whole numerator, so a one-term (**monomial**) denominator divides **every** term of the numerator: $\dfrac{a+b}{c} = \dfrac{a}{c} + \dfrac{b}{c}$. Dividing only one of the terms is a common error.

**Example 3**

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Divide each term of the numerator by the denominator, and simplify.

(a) $\dfrac{\sqrt{70}-\sqrt{42}}{\sqrt{7}}$

(b) $\dfrac{12\sqrt{15}+8\sqrt{6}}{4\sqrt{3}}$

(c) $\dfrac{6\sqrt{40}-9\sqrt{24}}{3\sqrt{2}}$

### 3. Rationalizing a Monomial Denominator

A radical expression is in **simplest form** when no radicand has a perfect-square (or perfect-cube, ...) factor, **no radical is in a denominator**, **no fraction is under a radical**, like radicals are combined and the fraction is reduced.

Removing the radical from a denominator is called **rationalizing the denominator**. Multiply the numerator and the denominator by the radical in the denominator; this multiplies by 1, so the value does not change. For $m\sqrt{b}$, multiply by $\sqrt{b}$ only (not by $m\sqrt{b}$). Simplify a radical in the denominator first: it keeps the numbers small. A fraction under a radical is split first: $\sqrt{\dfrac{a}{b}} = \dfrac{\sqrt{a}}{\sqrt{b}}$.

$$\frac{a}{\sqrt{b}} \cdot \frac{\sqrt{b}}{\sqrt{b}} = \frac{a\sqrt{b}}{b} \qquad \frac{a}{m\sqrt{b}} \cdot \frac{\sqrt{b}}{\sqrt{b}} = \frac{a\sqrt{b}}{mb} \qquad (b > 0)$$

**Example 4**

Rationalize the denominator and simplify.

(a) $\dfrac{5}{\sqrt{3}}$

(b) $\dfrac{\sqrt{2}}{\sqrt{7}}$

(c) $\dfrac{14}{3\sqrt{7}}$

(d) $\dfrac{10}{\sqrt{20}}$

(e) $\dfrac{-9}{2\sqrt{6}}$

(f) $\sqrt{\dfrac{5}{8}}$

**Example 5**

When the numerator has two terms, multiply *every* term by the radical. Reduce only by a factor that divides *every* term of the numerator and the denominator.

(a) Simplify $\dfrac{8\sqrt{15}-4\sqrt{6}}{2\sqrt{3}}$ by dividing each term.

(b) Simplify the same expression by rationalizing first.

(c) $\dfrac{4+\sqrt{6}}{\sqrt{2}}$

(d) $\dfrac{3\sqrt{5}-10}{2\sqrt{5}}$

**4. Cube-Root and Higher-Index Denominators**

To rationalize $\sqrt[3]{b}$, multiply by the cube root that makes the radicand a **perfect cube**. Multiplying by $\sqrt[3]{b}$ itself is not enough: $\sqrt[3]{2}\cdot\sqrt[3]{2}=\sqrt[3]{4}$ is still irrational, but $\sqrt[3]{2}\cdot\sqrt[3]{4}=\sqrt[3]{8}=2$. For a fourth root, make a perfect fourth power.

$$\frac{5}{\sqrt[3]{2}}\cdot\frac{\sqrt[3]{4}}{\sqrt[3]{4}}=\frac{5\sqrt[3]{4}}{\sqrt[3]{8}}=\frac{5\sqrt[3]{4}}{2}$$

Ask: “what times the radicand makes a perfect cube?”

**Example 6**

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Rationalize the denominator and simplify.

(a) $\dfrac{4}{\sqrt[3]{3}}$ (b) $\dfrac{10}{\sqrt[3]{25}}$ (c) $\dfrac{9}{2\sqrt[3]{6}}$

(d) $\sqrt[3]{\dfrac{3}{4}}$ (e) $\dfrac{12}{\sqrt[3]{18}}$ (f) $\dfrac{6}{\sqrt[4]{27}}$

### 5. Quotients with Variables

State the restrictions first. An even-index radical needs a radicand $\geq 0$, and a denominator cannot be 0, so a variable radicand in a denominator must be $> 0$ (even index) or $\neq 0$ (odd index). Then divide or rationalize exactly as with numbers.

**Example 7**

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Simplify. State the restrictions on the variable.

(a) $\dfrac{\sqrt{45x^5}}{\sqrt{5x}}$ (b) $\dfrac{\sqrt{14a^3}+\sqrt{21a}}{\sqrt{7a}}$ (c) $\dfrac{6}{\sqrt{x}}$

(d) $\dfrac{3x}{\sqrt{6x}}$ (e) $\sqrt{\dfrac{5}{8y}}$ (f) $\dfrac{4}{\sqrt[3]{2m}}$

### 6. Problems with Quotients

In a measurement problem, dividing by a radical is common: a length is an area divided by another length. Give exact answers in simplest form, with units.

**Example 8**

(a) The diagonal of a square is $\sqrt{2}$ times its side. A square has a diagonal of 12 cm. Determine the exact side length and the area.

(b) A rectangular banner has an area of $(6\sqrt{30} + 9\sqrt{10})$ $m^2$ and a height of $3\sqrt{5}$ m. Determine its length and its perimeter.

(c) Solve $x\sqrt{6} + \sqrt{8} = \sqrt{50}$. Write $x$ in simplest form.

## Lesson 6: Rationalizing Binomial Denominators with Conjugates

*Multiplying by the conjugate to clear a two-term denominator, then simplifying the result fully.*

### 1. Conjugates

Two binomials that differ only in the sign between their terms are **conjugates**: $a + \sqrt{b}$ and $a - \sqrt{b}$, or $\sqrt{a} + \sqrt{b}$ and $\sqrt{a} - \sqrt{b}$. Their product is a difference of squares (Lesson 4), so the middle terms cancel and every radical disappears: the product is **rational**.

$$(a + \sqrt{b})(a - \sqrt{b}) = a^2 - b \qquad (\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b}) = a - b \qquad (m\sqrt{a} + n\sqrt{b})(m\sqrt{a} - n\sqrt{b}) = m^2a - n^2b$$

**Example 1**

Write the conjugate of each binomial, and find the product of the binomial and its conjugate.

| Binomial | Conjugate | Product |
|---|---|---|
| $4 + \sqrt{3}$ | | |
| $\sqrt{7} - \sqrt{2}$ | | |
| $2\sqrt{5} + 3$ | | |
| $3\sqrt{2} - 2\sqrt{6}$ | | |

### 2. Rationalizing a Binomial Denominator

Multiplying by the radical alone does not work for a binomial: $\dfrac{1}{2+\sqrt{3}} \cdot \dfrac{\sqrt{3}}{\sqrt{3}} = \dfrac{\sqrt{3}}{2\sqrt{3}+3}$ still has a radical in the denominator. Instead, multiply the numerator and the denominator by the **conjugate of the denominator**.

**Steps:** (1) multiply the numerator and denominator by the conjugate of the denominator; (2) expand the numerator; (3) the denominator becomes rational (difference of squares); (4) simplify: write radicals in simplest form, collect like terms, reduce by a factor of *every* term, and keep the denominator positive.

**Example 2**

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Rationalize the denominator and simplify.

(a) $\dfrac{6}{3-\sqrt{5}}$

(b) $\dfrac{10}{\sqrt{7}+\sqrt{2}}$

(c) $\dfrac{\sqrt{3}}{2\sqrt{3}-3}$

(d) $\dfrac{4}{\sqrt{6}-2\sqrt{2}}$

### 3. Numerators with Two Terms

When the numerator is also a binomial, expand it with FOIL (or as a perfect square when the numerator is the same as the conjugate you multiply by), then simplify.

**Example 3**

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Rationalize the denominator and simplify.

(a) $\dfrac{\sqrt{5}+1}{\sqrt{5}-1}$

(b) $\dfrac{2\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$

(c) $\dfrac{\sqrt{6}+3}{\sqrt{6}-2}$

### 4. Simplifying Fully

Write every radical in simplest form *before* rationalizing; the conjugate is then smaller and common factors are easier to see. After rationalizing, look for a factor of every term that also divides the denominator.

**Example 4**

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Simplify.

(a) $\dfrac{\sqrt{6}}{\sqrt{18}+\sqrt{12}}$

(b) $\dfrac{6-\sqrt{12}}{\sqrt{3}-1}$

### 5. Binomial Denominators with Variables

State the restrictions: each even-index radicand must be $\geq 0$, and the denominator cannot be 0. For example, $\sqrt{x}-3=0$ when $x=9$, so $\dfrac{1}{\sqrt{x}-3}$ needs $x \geq 0$ and $x \neq 9$. The conjugate works the same way: $(\sqrt{x}-3)(\sqrt{x}+3)=x-9$.

**Example 5**

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Rationalize the denominator. State the restrictions on the variables.

(a) $\dfrac{3}{\sqrt{x}+2}$ (b) $\dfrac{2\sqrt{x}}{\sqrt{x}-3}$ (c) $\dfrac{\sqrt{a}-\sqrt{b}}{\sqrt{a}+\sqrt{b}}$

### 6. Mixed Operations and Problems

To add or subtract fractions with radical denominators, rationalize each fraction first, then collect like terms. Use the order of operations, and multiply or divide using the rules from Lessons 4 and 5.

**Example 6**

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Simplify.

(a) $(2\sqrt{3}-1)(\sqrt{3}+4)-\dfrac{6}{\sqrt{3}}$ (b) $\dfrac{1}{2+\sqrt{3}}+\dfrac{1}{2-\sqrt{3}}$ (c) $\dfrac{3}{\sqrt{5}}-\dfrac{2}{\sqrt{5}+1}$

**Example 7**

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(a) The **golden ratio** is $\varphi=\dfrac{1+\sqrt{5}}{2}$. Show that $\dfrac{1}{\varphi}=\varphi-1$.

(b) A rectangle has an area of 11 cm$^2$ and a width of $(\sqrt{15}-2)$ cm. Determine its exact length and perimeter.

# Lesson 7: Unit 2 Review

*Every rule of the unit on one page, mixed examples that combine them, and a review assignment written like the unit test.*

## 1. Unit Summary

In $5\sqrt[3]{2}$ the **index** is 3, the **radicand** is 2 and the coefficient is 5. A **mixed radical** has a coefficient other than 1 ($5\sqrt[3]{2}$); an **entire radical** has coefficient 1 ($\sqrt[3]{250}$). No calculator is used in this unit: every answer is exact and in **simplest form**.

| Skill | Rule | Example |
|---|---|---|
| Mixed → entire | Raise the coefficient to the power of the index and multiply it into the radicand: $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. | $3\sqrt[3]{2} = \sqrt[3]{27 \cdot 2} = \sqrt[3]{54}$ |
| Entire → mixed | Take out the largest perfect $n$th-power factor: $\sqrt[n]{a^n b} = a\sqrt[n]{b}$. | $\sqrt{72} = \sqrt{36}\sqrt{2} = 6\sqrt{2}$ |
| Compare and order | Write each number as an entire radical with the same index; the larger radicand is the larger number. | $4\sqrt{3} = \sqrt{48}$ and $3\sqrt{5} = \sqrt{45}$, so $3\sqrt{5} < 4\sqrt{3}$ |
| Restrictions | Even index: radicand $\geq 0$. Odd index: any real number. A radical in a denominator also cannot equal 0. | $\sqrt{2x-6}$: $2x - 6 \geq 0$, so $x \geq 3$ |
| Variable radicands | Divide each exponent by the index: the quotient goes outside, the remainder stays inside. | $\sqrt{20x^5} = 2x^2\sqrt{5x}$, $x \geq 0$ |
| Add, subtract | Simplify every radical first, then combine **like radicals** (same index and same radicand) by adding their coefficients. | $\sqrt{12} + \sqrt{27} = 2\sqrt{3} + 3\sqrt{3} = 5\sqrt{3}$ |
| Multiply | Same index: $m\sqrt[n]{a} \cdot k\sqrt[n]{b} = mk\sqrt[n]{ab}$. Distribute, or multiply binomials term by term; $(\sqrt{a})^2 = a$. | $2\sqrt{3} \cdot 5\sqrt{6} = 10\sqrt{18} = 30\sqrt{2}$ |
| Divide | $\dfrac{m\sqrt[n]{a}}{k\sqrt[n]{b}} = \dfrac{m}{k}\sqrt[n]{\dfrac{a}{b}}$. Divide each term of a numerator by a monomial denominator. | $\dfrac{6\sqrt{40}}{2\sqrt{5}} = 3\sqrt{8} = 6\sqrt{2}$ |
| Rationalize a monomial denominator | Multiply top and bottom by the radical that makes the radicand in the denominator a perfect power. | $\dfrac{3}{\sqrt{6}} = \dfrac{3\sqrt{6}}{6} = \dfrac{\sqrt{6}}{2}$<br>$\dfrac{4}{\sqrt[3]{2}} = \dfrac{4\sqrt[3]{4}}{\sqrt[3]{8}} = 2\sqrt[3]{4}$ |
| Rationalize a binomial denominator | Multiply top and bottom by the **conjugate** (same two terms, middle sign changed): $(a+\sqrt{b})(a-\sqrt{b}) = a^2 - b$. | $\dfrac{2}{3-\sqrt{7}} = \dfrac{2(3+\sqrt{7})}{9-7} = 3 + \sqrt{7}$ |

**Simplest form checklist.** (1) No perfect-power factor left in any radicand. (2) No fraction under a radical. (3) No radical in a denominator. (4) Like radicals combined and fractions reduced. (5) Restrictions stated for variables (we keep variables non-negative so that $\sqrt{x^2} = x$).

**Rational and irrational.** A **rational** number can be written as a fraction of two integers (its decimal ends or repeats). $\sqrt[n]{a}$ is rational only when $a$ is a perfect $n$th power; otherwise it is **irrational** (its decimal never ends or repeats), which is why we keep the radical as the exact answer. A rational number plus or times a non-zero irrational number is irrational, but two irrational numbers can give a rational result: $\sqrt{2} \cdot \sqrt{8} = 4$, and a binomial times its conjugate is always rational.

**Common errors.** $\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$, $(\sqrt{a} + \sqrt{b})^2 = a + 2\sqrt{ab} + b$ (not $a + b$), and only like radicals combine: $\sqrt{2} + \sqrt{3} \neq \sqrt{5}$.

## 2. Converting, Comparing and Restrictions

### Example 1

Without a calculator, arrange each set from least to greatest.

(a) $4\sqrt{3},\ 3\sqrt{5},\ 2\sqrt{13},\ 7,\ 5\sqrt{2}$

(b) $2\sqrt[3]{7},\ 3\sqrt[3]{2},\ \sqrt[3]{60},\ 4$

### Example 2

(a) State the restriction on the variable for which each radical is defined (a real number).

(i) $\sqrt{3x+12}$ (ii) $\sqrt{10-4x}$ (iii) $\sqrt[3]{x-9}$ (iv) $\dfrac{5}{\sqrt{x-1}}$

(b) Simplify. Assume $a, b, x, y \geq 0$.

(i) $\sqrt{75x^4y^7}$ (ii) $\sqrt[3]{-40a^7b^3}$ (iii) Write $3x\sqrt{2x}$ as an entire radical.

## 3. Adding, Subtracting and Multiplying

### Example 3

Simplify. Assume $x \geq 0$.

(a) $2\sqrt{50} - 3\sqrt{18} + \sqrt{32}$ (b) $\sqrt[3]{54} + 4\sqrt[3]{16} - \sqrt[3]{250}$ (c) $\sqrt{12x^3} - x\sqrt{27x} + 5\sqrt{3x^3}$

### Example 4

Expand and simplify.

(a) $3\sqrt{2}\left(4\sqrt{6}-\sqrt{10}\right)$

(b) $\left(2\sqrt{3}-\sqrt{5}\right)\left(\sqrt{3}+4\sqrt{5}\right)$

(c) $\left(3\sqrt{2}-2\right)^2$

(d) $\left(5+2\sqrt{6}\right)\left(5-2\sqrt{6}\right)$

**4. Dividing and Rationalizing**

**Example 5**

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Simplify. Write each answer with a rational denominator.

(a) $\dfrac{10\sqrt{42}}{5\sqrt{6}}$

(b) $\dfrac{\sqrt{45}-6\sqrt{10}}{3\sqrt{5}}$

(c) $\dfrac{15}{2\sqrt{3}}$

(d) $\dfrac{6}{\sqrt[3]{9}}$

(e) $\sqrt{\dfrac{5}{8}}$

**Example 6**

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Rationalize the denominator and simplify.

(a) $\dfrac{8}{\sqrt{7}-\sqrt{3}}$

(b) $\dfrac{3+\sqrt{5}}{3-\sqrt{5}}$

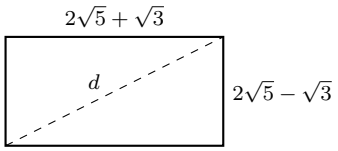
(c) $\dfrac{\sqrt{6}}{2\sqrt{3}+\sqrt{2}}$

## 5. Problem Solving

### Example 7

A rectangle has length $\left(2\sqrt{5}+\sqrt{3}\right)$ m and width $\left(2\sqrt{5}-\sqrt{3}\right)$ m.

(a) Determine the exact perimeter.




(b) Determine the exact area.

(c) Determine the exact length of the diagonal, $d$.

### Example 8

(a) Classify each number as rational or irrational.

(i) $\sqrt{196}$ (ii) $\sqrt[3]{-27}$ (iii) $\sqrt{40}$

(iv) $(\sqrt{7}-\sqrt{2})(\sqrt{7}+\sqrt{2})$ (v) $\dfrac{\sqrt{50}}{\sqrt{2}}$ (vi) $2+\sqrt{3}$

(b) The expression $\dfrac{4}{\sqrt{5}-1}-\sqrt{20}$ can be written in the form $a+b\sqrt{5}$. Determine $a$ and $b$.